Monash University, Faculty of Information Technology

FIT3080

Applied Class for Markov Decision Processes

Exercise 1: ***MDPs: Basic Conceptual Questions***

In the lecture, we discussed the Bellman Equation and how it can be used to characterize optimal utility in MDPs. For reference, recall this equation:

$$V^*(s) = \max_a \sum_{s'} T\,(s, a, s')\,[R(s, a, s') + \gamma V^*(s')]$$

a) What do we call γ in this equation? Why is this parameter required? What happens if γ becomes a very large or a very small value?
b) What are the major differences between the value iteration and policy iteration algorithms, and when might you prefer one to the other?
c) When does policy iteration end? Immediately after policy iteration ends (without performing additional computation), do we have the values of the optimal policy?
d) What changes if during policy iteration, you only run one iteration of Bellman update instead of running it until convergence? Do you still get an optimal policy?

Exercise 2: ***Modified Vacuum Cleaner World***

We modify the vacuum cleaner world example such that after any action, one of the two blocks will become randomly dirty with the probabilities p_L and p_R for the left and right blocks, respectively (note that $p_L + p_R = 1$). Using two actions, 1) staying in the current block and vacuum (abbreviated by "S&V") or 2) move to another block and vacuum (abbreviated by "M&V"), the vacuum cleaner should clean the dirt and receive a reward. The reward for staying in the same block and vacuum, "S&V", (if dirty) is the inverse of the probabilities and the reward for moving to other block and vacuum, "M&V", (if dirty) is equal to the ratio between the probabilities of target and source block probabilities. You will receive zero reward and the game ends if the block where the vacuum cleaner moves is not dirty. The goal is to maximise the accumulative sum of rewards.

a) Assuming you know the probabilities p_L and p_R, formulate this modified vacuum cleaner world as an MDP, writing down the reward function $R(s, a, s')$ and the transition function $T(s, a, s')$ as tables.
b) Assume the vacuum cleaner always start from the left block bL and the discount factor for the Bellman equation is equal to $\gamma = p_R$. Moreover, if at the beginning, the vacuum cleaner takes "M&V" action and receive a non-zero reward, it would be equal to 2. Calculate the exact values for p_L, p_R, γ, Transitions and Rewards.
c) Considering $V_0(s) = 0$ and starting from b_L, calculate the value iterations over the two iterations, $i = 2$.

Exercise 3: Value Iteration for the Maze (Python Code)

Recall the "Example: Value Iteration" from the week 8 seminar materials.

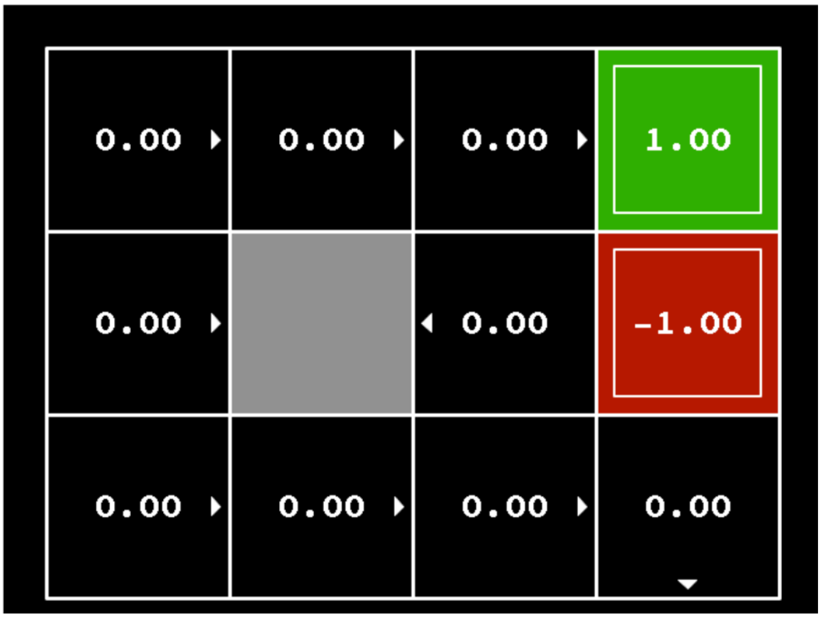

Figure 1: Maze Structure

In this exercise, you are required to implement the Value Iteration algorithm using the Bellman equation for the given maze (Fig. 1). The maze has two terminal states (i.e. green and red blocks), and there are no rewards on any other states. You need to write a Python code that performs the following tasks:

a) Implement the Value Iteration algorithm with a noise factor of 0.2 and a discount factor (γ) of 0.9.
b) Calculate the values and policy for each block of the maze.
c) Compare your calculated values and policy with those discussed in the lecture.